

Analyzing the Relationship Between Stellar Parameters and Flare Activity With TESS and Kepler

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Flare Detection with AltaiPony Python Algorithm

- The AltaiPony (AP) Python-based algorithm detrends, stores, and analyzes the long and short cadence TESS light curves to search for flares [1] [2]
- AP has a detrending algorithm that removes periodic and noise signals from light curves (Figure 1)
- The AP flare program has several limitations:
 - Blends – user must implement a separate algorithm to work with these
 - Pulsating stars – currently cannot distinguish pulsations from flares
 - Noise and incomplete detrending may result in false flare detections
- Over 1956 pairs of wide, noninteracting, TESS binaries were analyzed for flaring tendencies, with over 10% yielding some sort of flare signal
- ~200 flaring candidates were visibly non-contaminated

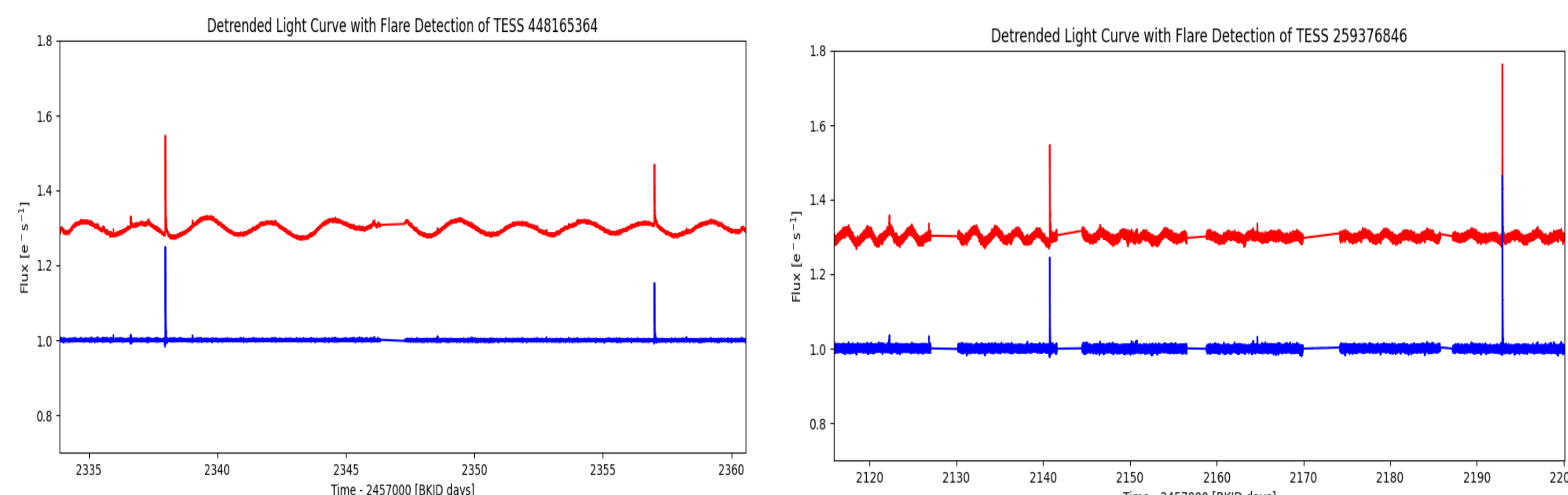


Figure 1 – Two TESS plots representing probable flare candidate examples. Both plots have the normalized detrended flux centered at 1.0 and normalized flux centered at 1.3.

How Does The Kepler Flare Detection Compare?

- AltaiPony was used on a sample of 393 well-vetted Kepler objects, where longer periods of observation were available
- In this sample, over 84% yielded a flare signature, with some having hundreds of flares captured in the four-year duration
- This percentage shows that longer durations of exposure can lead to higher chances of detecting a flare

Next Steps

- Analyze the Kepler Sample's flaring tendencies in the same manner as TESS
- Determine whether activity cycles can be detected

References

- [1] Davenport et al. (2019). The Evolution of Flare Activity with Stellar Age. The Astrophysical Journal, 871(241), 13. <https://doi.org/10.3847/1538-4357/aafb76>
- [2] Davenport, J. R. A. (2016). THE KEPLER CATALOG OF STELLAR FLARES. The Astrophysical Journal, 829(23), 12. <https://doi.org/10.3847/0004-637x/829/1/23>
- [3] Ilin et al. (2019). Flares in open clusters with K2. Astronomy & Astrophysics, 622(133), 16. <https://doi.org/10.1051/0004-6361/201834400>
- [4] Ilin et al. (2021). Flares in open clusters with K2. Astronomy & Astrophysics, 645(A42), 25. <https://doi.org/10.1051/0004-6361/202039198>
- [5] Wright et al. (2011). THE STELLAR-ACTIVITY-ROTATION RELATIONSHIP AND THE EVOLUTION OF STELLAR DYNAMOS. The Astrophysical Journal, 743(48), 16. <https://doi.org/10.1088/0004-637x/743/1/48>
- [6] Gruner, D. et al. (2023). New insights into the rotational evolution of near-solar age stars from the open cluster M 67. Astronomy and Astrophysics, 672(A159), 35. <https://doi.org/10.1051/0004-6361/202345942>

ABSTRACT

We present the results of our study with stellar flare frequency among lower main sequence stars using TESS (Transiting Exoplanet Survey Satellite) and Kepler Mission observations for which rotation periods are available. Target pixel files of 3912 wide, noninteracting binaries observed during TESS Cycle 3 were analyzed for flare activity using a Python code incorporating the AltaiPony Flare Detection Algorithm (Ilin, 2021). Additionally, Target Pixel Files of various well-vetted Kepler Mission objects were also analyzed for their flaring tendencies. The resulting flare frequencies and equivalent durations of the TESS objects were used to examine the correlation between rotation period and flare frequency. The well-known relation between stellar rotation and age, known as Gyrochronology, implies that as a star ages, its rotation period increases. We found through our study of TESS objects that flare rate decreases as stellar rotation periods increase, supporting the notion that flare activity diminishes with age. We are investigating whether the relationship between flare rate and rotation period observed in the TESS mission data is consistent across other datasets, specifically focusing on the Kepler Mission's observations. It is also of interest to see how TESS and Kepler data differ when detecting flare frequencies and which mission shows more clarifying trends.

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Stellar Activity Relationships with TESS

- The Rossby number (R_o) is defined as the ratio of rotation period and the convective turnover time [5]; it is more closely related to a star's activity level than rotation period alone
- Figure 2 displays the correlations we found between stellar activity, mass, luminosity, and the rotation periods (determined by the Lomb-Scargle periodogram analysis)
- Flare frequencies in these images were underestimated in value, but the trends and patterns are not affected
- Each plot below shows a decrease in flare frequency and flare intensity as rotation periods increase (aging stars)
- Kepler flares have not been analyzed like TESS flares, so these plots only represent TESS flaring tendencies*

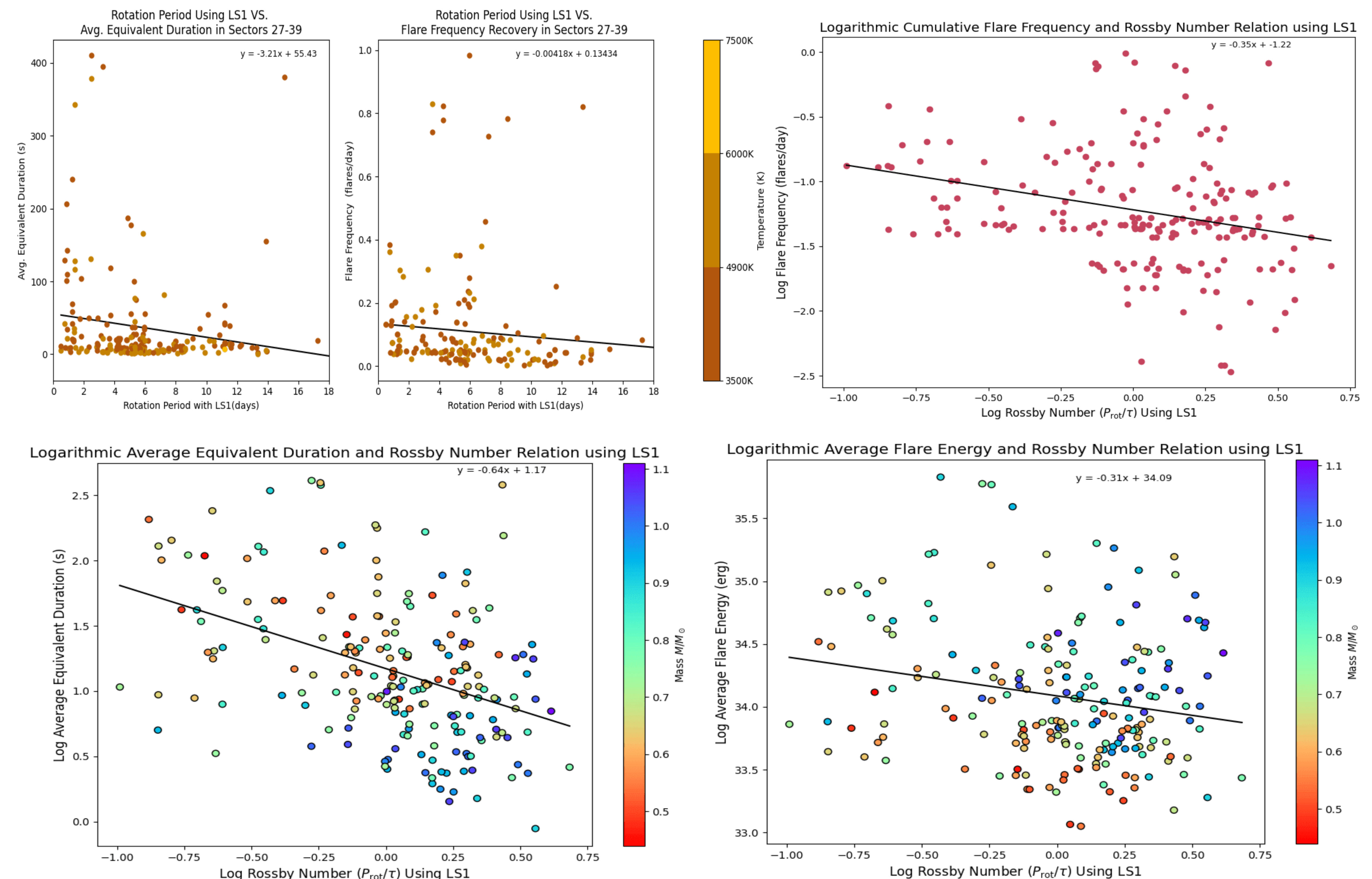


Figure 2 - Four plots showing different relationships between various parameters (such as mass, temperature, Rossby number and rotation periods) and stellar flare activity with TESS data. The top left plot contains a temperature color bar, and the bottom two plots contain mass color bars.